

Algorithms

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1. Matchings

1.1. Bipartite Maximum Matching

Definition 1.1.1 (matching): A *matching* in an undirected graph is an edge set such that every vertex belongs to either one (*matched vertex*) or $\emptyset$ (*free vertex*) edges of the matching.

Maximum Matching Problem: Given a graph G , find a matching with as many edges as possible.

Given a graph G and a suboptimal matching M , how should we improve it?

Definition 1.1.2 (M -augmenting path): Suppose G is a graph and M is a matching. Then, an M -augmenting path in G is a path P in G that starts and ends at free vertices and alternates between edges in M and not in M .

Lemma 1.1.1: If P is an M -augmenting path then $M \oplus P$ is a matching and

$$|M \oplus P| = |M| + 1.$$

Proof: Consider a vertex $v \in M \oplus P$. It suffices to prove that v is adjacent exactly one edge of $M \oplus P$. Note that $M \oplus P = (M \setminus P) \sqcup (P \setminus M)$. If $v \in M \setminus P$ then v is adjacent to exactly one edge in $M \setminus P$ as M is a matching. Similarly, if $v \in P \setminus M$ then there are two cases to consider. First, if v is free with respect to M , then v is an endpoint of P and hence, is only adjacent to one edge in P . Second, if v is matched with respect to M , then v must be an interior vertex of P as it isn't free. Thus, it must be adjacent to exactly one edge in M and one edge not in M . Consequently, v is adjacent to exactly one edge in $P \setminus M$.

Note that P alternates between edges in M and not in M and starts and ends with edges not in M . We have $|P \setminus M| = |M| + 1$ and $|M \cap P| = |M|$. Thus, $|M \oplus P| = |M| + 1$.

□

1.2. Hopcroft-Karp Algorithm

1.3. Min-cost Bipartite Perfect Matching, LP Relaxation

1.4. *Mise en Place* : Ski-Rental Problem

1.5. Online Matching

1.6. Algebraic Algorithms